Food-Dependent Differences in Foraging Path Length Between Rover and Sitter *Drosophila melanogaster* Larvae

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Introduction

The ability to find food is one of the most important aspects for survival. In *Drosophila melanogaster* (fruit flies) there are two dominant behaviors by which these flies find food: rovers, a behavior in which larvae are actively moving around to find food; and sitters, who find food once and stay at that food source (Sokolowski, 1980). It has been previously established that the difference in distance travelled between these two variants is dependent on the quantity of food (Kaun et al., 2007).

Furthermore, research has found that this behavior is determined by alleles on the fly's foraging gene (FOR) (de Belle et al., 1989). The two most commonly found alleles of this gene are for-R, which corresponds to Rovers; and for-S, which corresponds to sitters. When transcribed and translated, this gene produces protein kinase G, which when activated, changes how neurons in the fruit fly's brain fire therefore altering their behavior as a result (Osborne et al., 1997).

Path length was measured using the larval foraging assay, in which a single third-instar larva was placed at the center of a plate and the distance it travelled over five minutes was recorded (Sokolowski, 1980). This study aimed to verify these findings independently, hypothesizing that rover larvae would travel farther than sitter larvae in every substrate condition, and that both variants would travel shorter distances as the quantity of food available increased.

Materials and Methods

Materials

Assay plates were prepared using deionized (DI) water, agar powder, instant active dry yeast, a microwave and beaker for dissolving agar, a scale and weigh paper for massing dry reagents, Pasteur pipettes for transferring media, and 100 mm × 15 mm polystyrene petri dishes, which served as the arena for each individual larva. A paint brush was used to transfer larvae. Each trial was recorded from above with a smartphone camera held on a stand. A timer was used to keep the assay period at five minutes for every larva. To keep track of each container, a fine-tipped permanent marker was used to label plates and lids by genotype and substrate condition.

Tracking Software

Custom tracking software was written in Python 3.13 and built around trackpy 0.7, an established implementation of the Crocker–Grier particle-tracking algorithm. The program sampled one frame every 2 s (0.5 Hz) to track the larvae. Each plate was isolated either by automatic circle detection (Hough transform) or by manual selection, and pixel displacements were converted to millimeters using the known 100 mm plate diameter as the calibration standard. A per-pixel median of the sampled frames was computed to allow for the filtering out of the static background in each frame. Detections were linked into trajectories using a search radius corresponding to 4 mm s^{-1} of travel and a gap tolerance of 20 sampling intervals. Because larvae were hard to detect when pressed against the plate wall, gaps occurring at the wall were bridged along the rim arc. Path length was computed as the summed displacement along the reconstructed trajectory. The path length of each larva was taken from its trajectory, and a mean path length was calculated for each variant within each substrate condition. On the two-spot

plates, where five larvae of the same variant shared one plate, each path was traced by hand from the recording rather than taken from the automatic tracking output. Claude was used in the making of this software.

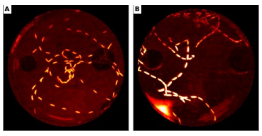


Figure 1. Reconstructed larval trajectories produced by the custom tracking software. Each panel shows the larvae over the dish over one five-minute trial, viewed from above. Panel (A) is a rover larva, whose path crossed a wider area and reached the plate wall, and panel (B) is a sitter larva, whose path stayed largely within the central region of the plate.

Animal Preparation

Each larva was visually verified to be alive and moving before testing, and each larva was tested only once. One rover larva and one sitter larva were tested on the agar-only and uniform yeast plates, and five larvae of each variant were tested on the two-spot plates.

Preparation of Substrate Conditions

Three substrate conditions were prepared. For all conditions, an agar base was made by combining approximately 15 mL DI water and 0.225 g agar powder per petri dish in a beaker, bringing the mixture to a light boil using a microwave with periodic stirring until the agar dissolved, removing it from the microwave, and pouring approximately 15 mL into each dish once bubbling had stopped. Plates were left uncovered until fully solidified. For the other two conditions a yeast paste was prepared by mixing 1 g dry yeast into 2 mL DI water per plate and stirring until it was smooth enough to pipette. On the uniform yeast plates the paste was applied dropwise across the whole agar surface and spread with a paint brush. On the two-spot plates two circles of paste were placed on opposite sides of the dish, leaving the rest of the agar surface bare.

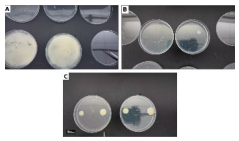


Figure 2. Assay plates for the three substrate conditions. All plates were 100 mm × 15 mm petri dishes holding approximately 15 mL of solidified agar. Panel (A) shows yeast-coated plates, (B) agar-only control plates, and (C) two-spot plates, each carrying two circles of yeast paste.

Larval Foraging Assay

Each larva was tested individually on the agar-only and uniform yeast plates, while five larvae of the same variant were run together on each two-spot plate. On the single-larva plates, the larva was placed at the center of its assay plate and a 5-minute timer was started once the larva began to move. At the end of the assay period, recording was stopped and the larva was euthanized via ethanol.

Data and Results

Substrate condition	Genotype	Individual path lengths (mm)	n	Mean path length (mm)	Measurement
Agar only (control)	Rover (for-R)	72.1	1	72.1	Automatic
Agar only (control)	Sitter (for-S)	62.2	1	62.2	Automatic

Substrate condition	Genotype	Individual path lengths (mm)	n	Mean path length (mm)	Measurement
Uniform yeast	Rover (for-R)	43.9	1	43.9	Automatic
Uniform yeast	Sitter (for-S)	1.0	1	1.0	Hand-scored
Two-spot yeast	Rover (for-R)	106.0, 246.6, 120.1, 149.6, 250.2	5	174.5	Hand-traced
Two-spot yeast	Sitter (for-S)	98.2, 83.6, 179.1, 90.6, 130.1	5	116.3	Hand-traced

Table 1. Individual and mean larval path lengths by genotype and substrate condition. Each value is the total distance travelled by one larva over a single five-minute trial, converted into millimeters using the 100 mm plate diameter as the calibration standard. Path lengths on the agar-only and uniform yeast plates were produced by the automatic tracking software. The sitter larva on the uniform yeast plate showed no locomotion, so its value was scored by hand. Path lengths on the two-spot plates were traced by hand because five larvae of the same variant shared each plate, which made automatic identification of each larva unreliable.

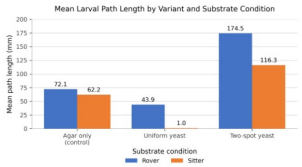


Figure 3. Mean path length of rover and sitter larvae across the three substrate conditions. Blue bars show rover larvae and orange bars show sitter larvae. Each bar is the mean distance travelled over a single five-minute trial, on agar-only control plates, plates coated in yeast, and plates carrying two yeast spots. The value above each bar gives that mean in millimeters, and the individual measurements behind every bar are listed in Table 1. Rover larvae travelled farther than sitter larvae in all three conditions, with the largest difference on the uniform yeast plates and the smallest on the agar-only control. Sample sizes were one larva per variant on the agar-only and uniform yeast plates and five larvae per variant on the two-spot plates.

Written Results

Rover larvae covered more distance than the sitter larvae in every condition (Table 1; Figure 3). On the agar-only control plate the rover larva travelled 72.1 mm and the sitter larva 62.2 mm. On the uniform yeast plate the rover larva travelled 43.9 mm and the sitter larva 1.0 mm. On the two-spot plates, where five larvae of each variant were tested, rover larvae travelled a mean of 174.5 mm and sitter larvae a mean of 116.3 mm. The sitter larva on the yeast plate did not move during its trial; only head movement was observed, and its path length was scored at 1.0 mm.

Path lengths had a wide range on the two-spot plates, with rover values ranging from 106.0 mm to 250.2 mm and sitter values from 83.6 mm to 179.1 mm (Table 1).

Both variants travelled the shortest distances on the uniform yeast plate and the longest distances on the two-spot plates, with the agar-only control plate falling between the two (Table 1; Figure 3).

Discussion

This experiment measured the distance travelled by rover and sitter larvae over five minutes on three substrates that differed in the amount of food available. The results supported part of the hypothesis. Rover larvae travelled farther than sitter larvae in all conditions, and both variants travelled the least on the uniform yeast plate, where food was most plentiful. The difference

between rovers and sitters was much smaller on the agar-only plate, which matches previous research showing that this difference becomes reduced or undetectable when food is absent (Sokolowski, 1980; de Belle et al., 1989). The second prediction, that both variants would travel shorter distances as the quantity of food increased, was not supported, because both travelled farthest on the two-spot plates even though those plates held less yeast than the uniform lawn. Path length therefore appears to depend on how the food was distributed rather than on how much of it was present, since a larva on a two-spot plate had to cross bare agar to reach either patch.

The difference between the two variants comes from the foraging gene, which encodes protein kinase G (Osborne et al., 1997). Rover larvae are known to carry the allele associated with higher PKG activity than sitter larvae. Because PKG production affects the circuits that control locomotion and feeding, a difference in the protein amount leads to a difference in behavior. Furthermore, PKG activity keeps a larva moving when it is eating food, so a rover continues cover more distance, while a sitter settles where it first encounters food and feeds in place.

One possible reason for this difference is the small sample size. Only one rover and one sitter were tested for the agar-only and uniform yeast conditions, meaning that one unusually active or inactive larva could have skewed the results. Differences in yeast thickness, agar moisture, larval age, and errors while tracking the larvae may have also affected the results. Future experiments should test more larvae in each condition and could change the amount or distribution of yeast to better determine how food availability affects rover and sitter behavior.

This experiment shows that a single gene can hold two behavioral strategies in one population. Moving a lot pays off when food is scarce, while staying in place pays off when food is abundant, so neither allele replaces the other.

References

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